

So let's get rid of the expression derivative d/dt and replace it with an appropriate functional derivative. If f is a function then we will write Df as the new function that is the derivative of f :⁵

$$(Df)(t) = \left. \frac{d}{dx} f(x) \right|_{x=t}.$$

To do this for the Lagrange equation we need to construct a function to take the derivative of.

Given a configuration-space path w , there is a standard way to make the state-space path. We can abstract this method as a mathematical function Γ :

$$\Gamma[w](t) = (t, w(t), \frac{d}{dt}w(t)).$$

Using Γ we can write:

$$\frac{d}{dt}((\partial_2 L)(\Gamma[w](t))) - (\partial_1 L)(\Gamma[w](t)) = 0.$$

If we now define composition of functions $(f \circ g)(x) = f(g(x))$, we can express the Lagrange equations entirely in terms of functions:

$$D((\partial_2 L) \circ (\Gamma[w])) - (\partial_1 L) \circ (\Gamma[w]) = 0.$$

The functions $\partial_1 L$ and $\partial_2 L$ are partial derivatives of the function L . Composition with $\Gamma[w]$ evaluates these partials with coordinates and velocities appropriate for the path w , making functions of time. Applying D takes the time derivative. The Lagrange equation states that the difference of the resulting functions of time must be zero. This statement of the Lagrange equation is complete, unambiguous, and functional. It is not encumbered with the particular choices made in expressing the Lagrangian. For example, it doesn't matter if the time is named t or τ , and it has an explicit place for the path to be tested.

This expression is equivalent to a computer program:⁶

⁵An explanation of functional derivatives is in Appendix B, page 202.

⁶The programs in this book are written in Scheme, a dialect of Lisp. The details of the language are not germane to the points being made. What is important is that it is mechanically interpretable, and thus unambiguous. In this book we require that the mathematical expressions be explicit enough